

Recall that in the previous exercise sheet you studied the flat FLRW metric,

$$ds^2 = -dt^2 + a^2(t) (dx^2 + dy^2 + dz^2), \quad (1)$$

which you found had

$$R_{tt} = -3\frac{\ddot{a}}{a}, \quad R_{ii} = 2\dot{a}^2 + a\ddot{a}, \quad (2)$$

with $i = x, y, z$ and $\cdot = \partial_t$.

a)* We now want to study a Universe filled with a perfect fluid,

$$T_{\mu\nu} = (\rho + p)U_\mu U_\nu + pg_{\mu\nu}, \quad (3)$$

where U^μ is the fluid's 4-velocity, $U^\mu U_\mu = -1$.

Argue that, if we want to keep homogeneity and isotropy, the fluid must be at rest with the FLRW geometry, $U^\mu = (1, 0, 0, 0)$. Show that this implies

$$T^\mu{}_\nu = \text{diag}(-\rho, p, p, p). \quad (4)$$

b)* Using the Einstein Equations and the conservation of the stress-energy tensor $\nabla^\mu T_{\mu\nu} = 0$, derive the Friedmann Equations for a Universe filled with a perfect fluid,

$$(F1): \quad \left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho, \quad (5)$$

$$(F2): \quad \frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p), \quad (6)$$

$$(F3): \quad \dot{\rho} = -3\frac{\dot{a}}{a}(\rho + p). \quad (7)$$

Argue that, if $\rho > 0$ and $p \geq 0$, there cannot be a static solution to the Friedmann Equations.

If the fluid were not at rest, there would be a preferred isotropy.

$$\begin{aligned} T_{\mu\nu} &= (\rho + p)U_\mu U_\nu + pg_{\mu\nu} \\ &= (\rho + p)\text{diag}(1, 0, 0, 0) + p\text{diag}(-1, 0, 0, 0) \\ &= \text{diag}(\rho, p, p, p) \end{aligned}$$

Then raising one index yields

$$T^\mu{}_\nu = \text{diag}(-\rho, p, p, p)$$

Einstein Equations (with cosmological constant)

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}$$

$$\text{EFE} \quad R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = 8\pi G T_{\mu\nu}$$

Since R , g and T are all diagonal, the only equations are diagonal

$$-3\frac{\ddot{a}}{a} - \frac{1}{2}\left[6\left(\frac{\ddot{a}}{a} + \frac{\dot{a}^2}{a^2}\right)\right](-1) = 8\pi G\rho$$

$$(2\dot{a}^2 + a\ddot{a}) - \frac{1}{2}\left[6\left(\frac{\ddot{a}}{a} + \frac{\dot{a}^2}{a^2}\right)\right]a^2 = 8\pi G\rho$$

Conservation law

$$\nabla^\mu T_{\mu\nu} = \nabla^\nu T_{\nu\nu} \text{ (no sum here)} \\ = 0$$

$$\begin{aligned} \nabla^\mu T_{\mu\nu} &= \nabla_\mu T^\mu{}_\nu \\ &= \partial_\mu T^\mu{}_\nu + \Gamma^\mu{}_{\nu\sigma} T^\sigma{}_\mu - \Gamma^\sigma{}_{\nu\mu} T^\mu{}_\sigma \end{aligned}$$

Recall the only nonzero Christoffels are Γ^0_{ii} and Γ^i_{i0} . Thus for $\nu=0$

$$\nabla^\mu T_{\mu 0} = \dot{\rho} + 3\frac{\dot{a}}{a}\rho + 3\frac{\dot{a}}{a}\rho = 0$$

For $\nu=1, 2, 3$, all terms in the equation vanish. In summary,

$$-3\frac{\ddot{a}}{a} - \frac{1}{2}\left[6\left(\frac{\ddot{a}}{a} + \frac{\dot{a}^2}{a^2}\right)\right](-1) = -3\frac{\ddot{a}}{a} + \frac{3\ddot{a}}{a} + 3\frac{\dot{a}^2}{a^2}$$

$$= 3\frac{\dot{a}^2}{a^2} = 8\pi G\rho$$

$$\frac{\dot{a}^2}{a^2} = \frac{8\pi G\rho}{3}$$

$$(2\dot{a}^2 + a\ddot{a}) - \frac{1}{2}\left[6\left(\frac{\ddot{a}}{a} + \frac{\dot{a}^2}{a^2}\right)\right]a^2 = (2\dot{a}^2 + a\ddot{a}) - 3\left(\frac{\ddot{a}}{a} + \frac{\dot{a}^2}{a^2}\right)a^2$$

$$= 2\dot{a}^2 + a\ddot{a} - 3a\ddot{a} - 3\dot{a}^2$$

$$= -\dot{a}^2 - 2a\ddot{a}$$

$$= 8\pi G\rho a^2$$

$$8\pi G\rho = -\left[\frac{\dot{a}^2}{a^2} + 2\frac{\ddot{a}}{a}\right]$$

$$= -\left[\frac{8\pi G\rho}{3} + 2\frac{\ddot{a}}{a}\right]$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}[\rho + 3p]$$

$$\dot{\rho} = -3 \frac{\dot{a}}{a} (\rho + p)$$

c)* Assuming that the fluid has a constant equation of state

$$w = p/\rho,$$

(8)

solve the Friedmann Equations, that is, find $\rho(a)$ and $a(t)$.

$$\frac{\dot{a}^2}{a^2} = \frac{8\pi G \rho}{3}$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3} [\rho + 3p]$$

$$\dot{\rho} = -3 \frac{\dot{a}}{a} (\rho + p)$$

First we eliminate p in favour of ρ

$$\frac{\dot{a}^2}{a^2} = \frac{8\pi G \rho}{3}$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G \rho}{3} (1+3w)$$

$$\dot{\rho} = -3 \frac{\dot{a}}{a} \rho (1+w)$$

Using the second equation to eliminate the first

$$\frac{\ddot{a}}{a} = -\frac{1}{2} \frac{\dot{a}^2}{a^2} (1+3w)$$

$$\ddot{a} = a \frac{d\dot{a}}{da}$$

$$a \frac{d\dot{a}}{da} = -\frac{1}{2} \frac{\dot{a}^2}{a} (1+3w)$$

$$\int \frac{1}{a} d\dot{a} = -\frac{1}{2} (1+3w) \int \frac{1}{a} da$$

$$\ln \dot{a} = -\frac{1}{2} (1+3w) \ln a + C$$

$$\dot{a} = A a^{-\frac{1}{2}(1+3w)}, A > 0$$

Integrating,

$$a^{\frac{1}{2}(1+3w)} da = A dt$$

$$a^{\frac{3}{2}(1+w)} = At + B, w \neq -1$$

By redefining $t=0$, we can remove B to find

$$a = A' t^{\frac{2}{1+3w}}$$

$$\dot{\rho} = -3 \frac{\dot{a}}{a} \rho (1+w)$$

$$= -3 \frac{\dot{a}}{a} \rho (1+w)$$

$$\frac{d}{dt} \ln \rho = -3(1+w) \frac{d}{dt} \ln a$$

$$\ln \rho = -3(1+w) \ln a + D$$

$$\rho = B a^{-3(1+w)}$$

d)* Analyze your solution for the three most common values for w : $w = 0$ for pressure-less matter, such as dust or cold dark matter (CDM); $w = 1/3$ for radiation, such as photons and ultra-relativistic particles like neutrinos; and $w = -1$ for dark energy (this can also be modelled by adding a cosmological constant Λ to the Einstein Equations).

What happens to $a(t)$ when $w = -1$?

Do these three kinds of fluids fail to satisfy any energy condition?

$$a = A' t^{\frac{2}{1+3w}}, w \neq -1$$

$$w = 0: \quad \begin{aligned} a &= A' t^{\frac{2}{3}} \\ \rho &= B a^{-3} \\ \alpha &= -2 \end{aligned}$$

Expansion decelerating

$$w = \frac{1}{3}: \quad \begin{aligned} a &= A' t^{1/2} \\ \rho &= B a^{-4} \\ \alpha &= -2 \end{aligned}$$

$$w = -1: a^{\frac{1}{2}(1+3w)} da = A dt$$

$$a^{-1} da = A dt$$

$$\ln a = A t + B$$

$$a = e^{A t + B}$$

$$\rho = \text{const.}$$

Recall for the energy conditions

NEC $\Rightarrow (\rho + p) \geq 0$,
 WEC $\Rightarrow \rho \geq 0$ and $(\rho + p) \geq 0$,
 DEC $\Rightarrow \rho \geq |p|$,
 SEC $\Rightarrow (\rho + 3p) \geq 0$ and $(\rho + p) \geq 0$.

$$p = w\rho$$

For $w = -1$ the SEC is violated, $w = 0$ and $w = \frac{1}{3}$ do not violate any energy conditions.

Numerous astronomical observations suggest that the Universe underwent a phase of accelerated ($\ddot{a} > 0$) expansion ($\dot{a} > 0$), called *inflation*, at its birth.

a)* Show that inflation is equivalent to requiring

$$-\dot{H}/H^2 < 1, \quad (9)$$

where $H = \dot{a}/a$ is the Hubble parameter.

Argue also that inflation can be modelled using a perfect fluid with

$$w < -1/3. \quad (10)$$

What energy condition do we need to violate achieve this?

$$\begin{aligned} \dot{H} &= \frac{\ddot{a}}{a} - \frac{\dot{a}^2}{a^2} \\ \frac{\dot{H}}{H^2} &= \left(\frac{\ddot{a}}{a} - \frac{\dot{a}^2}{a^2} \right) \left(\frac{a^2}{\dot{a}^2} \right) \\ &= \frac{\ddot{a}}{\frac{\dot{a}^2}{a}} - 1 \\ -\frac{\dot{H}}{H^2} < 1 &\Leftrightarrow -\frac{\ddot{a}}{\frac{\dot{a}^2}{a}} < 0 \end{aligned}$$

Since $a > 0$, this is accelerating if $\ddot{a} > 0$.

We recall from previously that $a^{\frac{3}{2}(1+w)}$ of

If $w < -\frac{1}{3}$, then $\frac{3}{2}(1+w) < 1$, thus $\dot{a} > 0$ and the universe is accelerating.

This violates the strong energy condition

b)* We now model inflation using a scalar field ϕ (the *inflaton*) with potential $V(\phi)$. We still assume homogeneity and isotropy, so the inflaton field can only be a function of time $\phi = \phi(t)$.

Identify the scalar field's energy density ρ_ϕ and pressure p_ϕ

$$\rho_\phi := -T^0_0, \quad (11)$$

$$p_\phi g^j_j := T^j_j, \quad (12)$$

from your results for the $T_{\mu\nu}$ of a scalar field from the previous exercise sheet.

Then, inserting these into the first two Friedmann equations, show

$$(F1): \quad H^2 = \frac{8\pi G}{3} \left(\frac{1}{2} \dot{\phi}^2 + V(\phi) \right), \quad (13)$$

$$(F2): \quad \dot{H} = -4\pi G \dot{\phi}^2. \quad (14)$$

Finally prove that the Klein-Gordon equation $\square\phi - V'(\phi) = 0$ yields

$$\ddot{\phi} + 3\frac{\dot{a}}{a}\dot{\phi} = -V'(\phi). \quad (15)$$

We recall that for a minimally coupled scalar field

$$T_{\mu\nu} = \nabla_\mu \phi \nabla_\nu \phi - \frac{1}{2} g_{\mu\nu} \nabla^\rho \phi \nabla_\rho \phi - g_{\mu\nu} V(\phi).$$

$$\begin{aligned} T^{\nu}_{\nu} &= g^{\nu\sigma} \nabla_\sigma \phi \nabla_\nu \phi - \frac{1}{2} g^{\nu\sigma} g_{\sigma\nu} \nabla^\rho \phi \nabla_\rho \phi - g^{\nu\sigma} g_{\sigma\nu} V(\phi) \\ &= g^{\nu\sigma} \nabla_\sigma \phi \nabla_\nu \phi - \frac{1}{2} \delta^{\nu}_{\nu} \nabla^\rho \phi \nabla_\rho \phi - \delta^{\nu}_{\nu} V(\phi) \\ T^0_0 &= -\dot{\phi}^2 - \frac{1}{2} (-\dot{\phi}^2) - V(\phi) \\ &= -\frac{1}{2} \dot{\phi}^2 - V(\phi) \end{aligned}$$

Hence $\rho_\phi = \frac{1}{2} \dot{\phi}^2 + V(\phi)$

Substituting directly yields $H^2 = \frac{8\pi G}{3} \left(\frac{1}{2} \dot{\phi}^2 + V(\phi) \right)$

$$T^1_1 = \frac{1}{2} \dot{\phi}^2 - V(\phi)$$

$$\begin{aligned} \dot{H} &= \frac{\ddot{a}}{a} - \frac{\dot{a}^2}{a^2} \\ &= -\frac{4\pi G}{3} (\rho + 3p) - H^2 \\ &= -\frac{4\pi G}{3} \rho - 4\pi G p - \frac{8\pi G}{3} \rho \\ &= -4\pi G \rho - 4\pi G p \\ &= -4\pi G \left[\frac{1}{2} \dot{\phi}^2 + V(\phi) + \frac{1}{2} \dot{\phi}^2 - V(\phi) \right] \\ &= -4\pi G \dot{\phi}^2 \end{aligned}$$

Klein-Gordon equation $\nabla_\mu \nabla^\mu \phi - V'(\phi) = 0$

$$\nabla_\mu \nabla^\mu \phi = \nabla_\mu (\partial^\mu \phi)$$

$$= \partial_\mu \partial^\mu \phi + \Gamma^\nu_{\mu\nu} \partial^\mu \phi$$

$$= -\ddot{\phi} - 3\frac{\dot{\phi}}{a}\dot{\phi}$$

Hence $\ddot{\phi} + 3\frac{\dot{\phi}}{a}\dot{\phi} = -V'(\phi)$

c)* Argue that, for inflation to happen, we need the potential energy V of the scalar field to be greater than its kinetic energy $\dot{\phi}^2$.

From a), we know that $\ddot{a} > 0$ iff $\rho + 3p < 0$

Substituting the expressions, this is equivalent to

$$\rho + 3p = \left(\frac{1}{2}\dot{\phi}^2 + V\right) + 3\left(\frac{1}{2}\dot{\phi}^2 - V\right) = 2\dot{\phi}^2 - 2V$$

Hence $\rho + 3p < 0$ iff $V > \dot{\phi}^2$

e) The FLRW metric can be generalized to

$$ds^2 = -dt^2 + a^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right] = -dt^2 + a^2(t) \left[d\chi^2 + f_k^2(\chi) d\Omega^2 \right], \quad (16)$$

where the parameter k describes the curvature of the spatial slices, and

$$f_k(\chi) = \begin{cases} \sin \chi & \text{for } k = 1, \\ \chi & \text{for } k = 0, \\ \sinh \chi & \text{for } k = -1. \end{cases} \quad (17)$$

If $k > 0$, the Universe is positively curved (a closed 3-sphere), if $k = 0$ the Universe is flat, and if $k < 0$ the Universe is negatively curved (an open 3-hyperboloid).

Moreover, we can add a cosmological constant Λ to the Einstein Equations. Then, the new Friedmann Equations for a Universe filled with a perfect fluid read

$$(F1): \quad \left(\frac{\dot{a}}{a} \right)^2 = \frac{\Lambda}{3} + \frac{8\pi G}{3} \rho - \frac{k}{a^2}, \quad (18)$$

$$(F2): \quad \frac{\ddot{a}}{a} = \frac{\Lambda}{3} - \frac{4\pi G}{3} (\rho + 3p), \quad (19)$$

$$(F3): \quad \dot{\rho} = -3 \frac{\dot{a}}{a} (\rho + p). \quad (20)$$

Feel free to rederive these results using `diffgeo` as an extra exercise.

Now, assuming that the Universe is filled only with pressure-less matter ($w = 0$), show that there exists a *static* solution ($\dot{a} = 0$ and $\ddot{a} = 0$) to the Friedmann Equations, with a critical density of exactly

$$\rho_c = \frac{\Lambda}{4\pi G}. \quad (21)$$

What must be the curvature of this Universe?

If $\ddot{a} = \dot{a} = 0$, and $w=0$, then $p=0$ and the equations reduce to

$$0 = \frac{\Lambda}{3} + \frac{8\pi G}{3} \rho - \frac{k}{a^2}$$

$$0 = \frac{\Lambda}{3} - \frac{4\pi G}{3} \rho$$

$$\dot{\rho} = 0$$

We simply solve the second equation for

$$\rho_c = \frac{\Lambda}{4\pi G}$$

The first equation yields

$$\begin{aligned} 0 &= \frac{\Lambda}{3} + \frac{8\pi G}{3} \frac{\Lambda}{4\pi G} - \frac{k}{a^2} \\ &= \Lambda - \frac{k}{a^2} \end{aligned}$$

Since $a^2 > 0$, we must have $k > 0$ for a solution.

f) Finally, perturb the previous solution $\rho = \rho_c + \delta\rho$, with $\delta\rho \ll \rho_c$, to show that this solution is in fact unstable: a small perturbation $\delta\rho$ induces a perturbation δa that will exponentially grow in time. Einstein's fix did not work, the Universe *must* be dynamical.

$$H^2 = \frac{\Lambda}{3} + \frac{8\pi G}{3} (\rho_c + \delta\rho) - \frac{k}{a^2}$$

$$= \Lambda + \frac{8\pi G}{3} \delta\rho - \frac{k}{a^2}$$

$$\frac{\ddot{a}}{a} = \frac{\Lambda}{3} - \frac{4\pi G}{3} (\rho_c + \delta\rho)$$

$$= -\frac{4\pi G}{3} \delta\rho$$

$$\delta\dot{\rho} = -3 \frac{\dot{a}}{a} (\rho_c + \delta\rho)$$

$$\approx -3 \frac{\dot{a}}{a} \rho_c$$

Use the Tolman-Oppenheimer-Volkoff equations for dm/dr and $d\Phi/dr$ to eliminate ρ and p from the equation for dp/dr . Show that the resulting equation can be written as

$$(r^{-1}e^{-\Psi}\zeta')' = (mr^{-3})'e^{\Psi}\zeta, \quad (22)$$

where

$$e^{2\Psi} := \left(1 - \frac{2m}{r}\right)^{-1}, \quad \zeta := e^{\Phi}, \quad (23)$$

and the primes denote d/dr derivatives.

$$(r^{-1}e^{-\Psi}\zeta')' = -\frac{1}{r^2}e^{-\Psi}\zeta' - \Psi'r^{-1}e^{-\Psi}\zeta' + r^{-1}e^{-\Psi}\zeta''$$

$$(mr^{-3})'e^{\Psi}\zeta = m'r^{-3}e^{\Psi}\zeta - 3mr^{-4}e^{\Psi}\zeta$$

$$\Phi'\zeta = \zeta' = \Phi'e^{\Phi}, \quad \zeta'' = (\Phi'^2 + \Phi'')e^{\Phi}$$

$$-\frac{1}{r^2}\Phi' - \Psi'r^{-1}\Phi' + r^{-1}(\Phi'^2 + \Phi'') = (m'r^{-3} - 3mr^{-4})e^{\Psi}$$

$$\Phi'' + \Phi'^2 - \left(\frac{1}{r} + \Psi'\right)\Phi' = re^{2\Psi}(m'r^{-3} - 3mr^{-4})$$

$$= \frac{r}{1 - \frac{2m}{r}} \left[4\pi r^2 \rho r^{-3} - 3mr^{-4} \right]$$

$$\Phi'' + \Phi'^2 - \left(\frac{1}{r} + \Psi'\right)\Phi' \underset{\substack{\uparrow \\ \text{to show}}}{=} \frac{1}{r^2(r-2m)} \left[4\pi r^3 \rho - 3m \right]$$

$$\text{In}[16] := \Psi[r_] := -\frac{1}{2} \log \left[1 - \frac{2m[r_]}{r} \right]$$

$$\text{In}[17] := \phi p[r_] := \frac{m[r_] + 4\pi r^3 p[r_]}{r(r-2m[r_])}$$

$$\text{In}[18] := \text{Simplify}[\phi p'[r_]]$$

$$\text{Out}[18] = \frac{1}{r^2(r-2m[r])^2} (2m[r]^2 + r^2 m'[r] + 4\pi r^4 p[r] (1 + 2m'[r]) + 4\pi r^5 p'[r] - 2m[r] (r + 8\pi r^3 p[r] + 4\pi r^4 p'[r]))$$

$$\text{In}[19] := \text{Simplify}[\phi p'[r_] /. \{p'[r] \rightarrow -(\rho[r] + p[r]) \phi p[r], m'[r] \rightarrow 4\pi r^2 \rho[r]\}]$$

$$\text{Out}[19] = \frac{1}{r^2(r-2m[r])^2} (2m[r]^2 - 2m[r] (r + 10\pi r^3 p[r] + 2\pi r^3 \rho[r]) + 4\pi r^4 (p[r] - 4\pi r^2 p[r]^2 + \rho[r] + 4\pi r^2 p[r] \times \rho[r]))$$

$$\text{In}[20] := \Psi'[r] /. \{p'[r] \rightarrow -(\rho[r] + p[r]) \phi p[r], m'[r] \rightarrow 4\pi r^2 \rho[r]\}$$

$$\text{Out}[20] = -\frac{\frac{2m[r]}{r^2} - 8\pi r \rho[r]}{2(1 - \frac{2m[r]}{r})}$$

$$\Phi'' + \Phi'^2 - \left(\frac{1}{r} + \Psi'\right)\Phi'$$

$$\text{In}[21] := \text{Simplify} \left[\frac{1}{r^2(r-2m[r])^2} (2m[r]^2 - 2m[r] (r + 10\pi r^3 p[r] + 2\pi r^3 \rho[r]) + 4\pi r^4 (p[r] - 4\pi r^2 p[r]^2 + \rho[r] + 4\pi r^2 p[r] \times \rho[r])) + \right.$$

$$\left. \phi p[r]^2 - \left(\frac{1}{r} - \frac{\frac{2m[r]}{r^2} - 8\pi r \rho[r]}{2(1 - \frac{2m[r]}{r})} \right) \phi p[r] \right]$$

$$= \text{Out}[21] = \frac{-3m[r] + 4\pi r^3 \rho[r]}{r^2(r-2m[r])}$$

Hence the equations are equivalent to TOV.

b) Use $d\rho/dr \leq 0$ to argue that $(mr^{-3})' \leq 0$.

$$(mr^{-3})' = m'r^{-3} - 3mr^{-4}$$

$$\text{Since } m' = 4\pi r^2 \rho,$$

$$(mr^{-3})' = \frac{4\pi r^2 \rho - 3m}{r^4}$$

$$\text{Now } m(r) = \int_0^r 4\pi \xi^2 \rho(\xi) d\xi$$

$$\leq \int_0^r 4\pi \xi^2 \rho(0) d\xi$$

$$= \frac{4}{3} \pi r^3 \rho(0)$$

$$\text{Hence } 4\pi r^2 \rho(r) - 3m(r) \leq 4\pi r^2 \rho(0) - 3m(r) = 0$$

c) It follows that $(r^{-1}e^{-\Psi}\zeta')' \leq 0$. By integrating this equation from r_1 to r and using the equation for $d\Phi/dr$, show that, for any $r_1 \leq r$,

$$\zeta'(r_1) \geq \left(1 - \frac{2m(r)}{r}\right)^{-1/2} \left(\frac{m(r)}{r^3} + 4\pi\rho(r)\right) \zeta(r) r_1 \left(1 - \frac{2m(r_1)}{r_1}\right)^{-1/2}. \quad (24)$$

$$(r^{-1}e^{-\Psi}\zeta')' \leq 0$$

$$\int_{r_1}^r (\zeta' e^{-\Psi} r^{-1})' d\xi = r^{-1} e^{-\Psi} \zeta' - r_1^{-1} e^{-\Psi(r_1)} \zeta'(r_1) \leq 0$$

$$\zeta'(r_1) \geq r_1 e^{\Psi(r_1)} r^{-1} e^{-\Psi} \zeta'$$

$$e^{2\Psi} = \left(1 - \frac{2m}{r}\right)^{-1}$$

$$\zeta' = \frac{\zeta'}{\zeta} \zeta$$

$$\frac{d\Phi}{dr} = \frac{m + 4\pi r^3 \rho}{r(r - 2m)}$$

$$\zeta'(r_1) \geq r_1 \left(1 - \frac{2m}{r_1}\right)^{-1/2} \frac{1}{r} \zeta(r) \left(1 - \frac{2m}{r}\right)^{1/2} \frac{m + 4\pi r^3 \rho}{r(r - 2m)}$$

$$= r_1 \left(1 - \frac{2m}{r_1}\right)^{-1/2} \frac{1}{r^3} \zeta(r) \left(1 - \frac{2m}{r}\right)^{1/2} (m + 4\pi r^3 \rho) \left(1 - \frac{2m}{r}\right)^{-1}$$

$$= r_1 \left(1 - \frac{2m}{r_1}\right)^{-1/2} \zeta(r) \left(1 - \frac{2m}{r}\right)^{1/2} \left(\frac{m}{r^3} + 4\pi\rho\right)$$

d) Now, $(mr^{-3})' \leq 0$ implies $m(r_1)/r_1 \geq m(r)r_1^2/r^3$ for $r_1 \leq r$. Use this to show

$$\int_0^r r_1 \left(1 - \frac{2m(r_1)}{r_1}\right)^{-1/2} dr_1 \geq \frac{r^3}{2m(r)} \left[1 - \left(1 - \frac{2m(r)}{r}\right)^{1/2}\right]. \quad (25)$$

$$\frac{m(r_1)}{r_1} \geq \frac{m(r)r_1^2}{r^3}$$

$$1 - \frac{2m(r_1)}{r_1} \leq 1 - \frac{2m(r)r_1^2}{r^3}$$

$$\left(1 - \frac{2m(r_1)}{r_1}\right)^{-1/2} \geq \left(1 - \frac{2m(r)r_1^2}{r^3}\right)^{-1/2}$$

$$\int_0^r r_1 \left(1 - \frac{2m(r_1)}{r_1}\right)^{-1/2} dr_1 \geq \int_0^r r_1 \left(1 - \frac{2m(r)r_1^2}{r^3}\right)^{-1/2} dr_1$$

Evaluate the integral on the right first with $u = r_1^3$, $du = 3r_1^2 dr_1$

$$\int_0^r r_1 \left(1 - \frac{2m(r)r_1^2}{r^3}\right)^{-1/2} dr_1 = \frac{1}{3} \int_0^{r^3} \left(1 - \frac{2mu}{r^3}\right)^{-1/2} du$$

$$\begin{aligned}
&= \frac{1}{2} \left[\left(1 - \frac{2m}{r^3} \right)^{1/2} \left(-\frac{2r^3}{2m} \right) \right]_0^{r_1} \\
&= -\frac{r^3}{2m} \left[\left(1 - \frac{2m}{r^3} \right)^{1/2} \right]_0^{r_1} \\
&= \frac{r^3}{2m} \left[1 - \left(1 - \frac{2m}{r^3} \right)^{1/2} \right] \\
&= \frac{r^3}{2m} \left[1 - \left(1 - \frac{2m}{r^3} \right)^{1/2} \right]
\end{aligned}$$

$$\int_0^{r_1} \left(1 - \frac{2m(r)}{r^3} \right)^{-1/2} dr \geq \frac{r^3}{2m} \left[1 - \left(1 - \frac{2m}{r^3} \right)^{1/2} \right]$$

e) Deduce that

$$\zeta(0) \leq \zeta(r) \left\{ 1 - \left(\frac{1}{2} + \frac{2\pi^2 p(r)}{m(r)} \right) \left[\left(1 - \frac{2m(r)}{r^3} \right)^{-1/2} - 1 \right] \right\}. \quad (26)$$

$$\int_0^r \frac{dr_1}{r_1} \geq \left(1 - \frac{2m(r)}{r^3} \right)^{-1/2} \left(\frac{m(r)}{r^3} + 4\pi p(r) \right) \zeta(r) \int_0^r r_1 \left(1 - \frac{2m(r_1)}{r_1^3} \right)^{-1/2} dr_1$$

$$\zeta(r) - \zeta(0) \geq \left(1 - \frac{2m}{r^3} \right)^{-1/2} \left(\frac{m}{r^3} + 4\pi p \right) \zeta(r) \frac{r^3}{2m} \left[1 - \left(1 - \frac{2m}{r^3} \right)^{1/2} \right]$$

$$\begin{aligned}
\zeta(0) &\leq \zeta(r) - \left(1 - \frac{2m}{r^3} \right)^{-1/2} \left(\frac{m}{r^3} + 4\pi p \right) \zeta(r) \frac{r^3}{2m} \left[1 - \left(1 - \frac{2m}{r^3} \right)^{1/2} \right] \\
&= \zeta(r) \left\{ 1 - \left(\frac{1}{2} + \frac{2\pi p r^3}{m} \right) \left[\left(1 - \frac{2m}{r^3} \right)^{-1/2} - 1 \right] \right\}
\end{aligned}$$

f) By definition $\zeta(r)$ is positive everywhere. In particular, $\zeta(0) > 0$. Deduce that

$$\frac{m(r)}{r} \leq \frac{2}{9} (1 - 6\pi r^2 p(r) + \sqrt{1 + 6\pi r^2 p(r)}).$$

Obviously

$$0 < \zeta(r) \left\{ 1 - \left(\frac{1}{2} + \frac{2\pi p r^3}{m} \right) \left[\left(1 - \frac{2m}{r^3} \right)^{-1/2} - 1 \right] \right\}$$

$$0 < 1 - \left(\frac{1}{2} + \frac{2\pi p r^3}{m} \right) \left[\left(1 - \frac{2m}{r^3} \right)^{-1/2} - 1 \right]$$

$$\left(\frac{1}{2} + \frac{2\pi p r^3}{m} \right) \left[\left(1 - \frac{2m}{r^3} \right)^{-1/2} - 1 \right] < 1$$

$$\left(1 - \frac{2m}{r^3} \right)^{-1/2} < \left(\frac{1}{2} + \frac{2\pi p r^3}{m} \right)^{-1} + 1$$

$$1 - \frac{2m}{r^3} > \left[\left(\frac{1}{2} + \frac{2\pi p r^3}{m} \right)^{-1} + 1 \right]^2$$

$$\text{In[25]:= Together} \left[\frac{1}{\left(\frac{1}{\frac{1}{2} + \frac{2\pi p r^3}{m}} + 1 \right)^2} \right]$$

$$\text{Out[25]=} \frac{(m + 4 \pi p r^3)^2}{(3 m + 4 \pi p r^3)^2}$$

$$= \left(\frac{m + 4\pi p r^3}{3m + 4\pi p r^3} \right)^2$$

$$\left(1 - \frac{2m}{r^3} \right) (3m + 4\pi p r^3)^2 > (m + 4\pi p r^3)^2$$

$$\left(1 - \frac{2m}{r^3} \right) (3m + 4\pi p r^3)^2 - (m + 4\pi p r^3)^2 > 0$$

$$\text{In[29]:= Collect} \left[\text{Expand} \left[\left(1 - \frac{2m}{r} \right) (3m + 4\pi p r^3)^2 - (m + 4\pi p r^3)^2 \right], m \right]$$

$$\text{Out[29]=} -\frac{18m^3}{r} + m^2 (8 - 48\pi p r^2) + m (16\pi p r^3 - 32\pi^2 p^2 r^5)$$

$$-\frac{18m^3}{r} + m^2 (8 - 48\pi p r^2) + m (16\pi p r^3 - 32\pi^2 p^2 r^5) > 0$$

$$-\frac{18m^2}{r} + m (8 - 48\pi p r^2) + (16\pi p r^3 - 32\pi^2 p^2 r^5) > 0$$

Since this is a quadratic inequality, we find the roots

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In[32]:= Solve[-(18 m^2)/r + m (8 - 48 p π r^2) + (16 p π r^3 - 32 p^2 π^2 r^5) == 0, m]
Out[32]= {{m -> (2/9) (r - 6 p π r^3 - r sqrt(1 + 6 p π r^2))}, {m -> (2/9) (r - 6 p π r^3 + r sqrt(1 + 6 p π r^2))}}
```

Since the sign of m^2 is negative,

$$m < \frac{2}{9} (r - 6 p \pi r^3 + r \sqrt{1 + 6 p \pi r^2})$$

$$\frac{m(r)}{r} < \frac{2}{9} (1 - 6 p \pi r^2 + \sqrt{1 + 6 p \pi r^2})$$

g) Find Buchdahl's inequality,

$$\frac{M}{R} < \frac{4}{9}, \quad (28)$$

where M and R are the total energy and radius of the star.

At the surface, $r=R$, $m(R)=M$, $p(R)=0$

$$\frac{M}{R} < \frac{2}{9} (1 - 6 p R^2 + \sqrt{1 + 6 p R^2})$$

$$= \frac{4}{9}$$

A greatly simplified but instructive solution to the TOV equations is found by assuming

$$\rho(r) = \begin{cases} \rho_0 & r < R, \\ 0 & r > R, \end{cases} \quad (29)$$

where $\rho_0 > 0$ is some positive constant and R is the radius of the star. That is, we are considering a star made from incompressible matter.

a)* Integrate TOV1 to find $m(r)$. Identify the total energy of the star M .

$$\frac{dm}{dr} = 4\pi r^2 \rho(r)$$

$$m(r) = \int_0^r 4\pi \xi^2 \rho(\xi) d\xi$$

$$= \begin{cases} \frac{4}{3} \pi r^3 \rho_0, & r \leq R \\ \frac{4}{3} \pi R^3 \rho_0, & r > R \end{cases}$$

b)* Define $x := p/\rho_0$ to write TOV3 as

$$\frac{dx}{dr} = -(1+x)(1+3x) \frac{4\pi \rho_0 r}{3 - 8\pi \rho_0 r^2}.$$

$$\frac{dp}{dr} = -(p + \rho) \frac{(m + 4\pi r^3 p)}{r(r - 2m)}$$

$$\frac{dx}{dr} = \frac{1}{\rho_0} \frac{dp}{dr}$$

$$\frac{dx}{dr} = -\left(\frac{p}{\rho_0} + 1\right) \frac{m + 4\pi r^3 p}{r(r - 2m)}$$

$$= -(x + 1) \rho_0 \frac{\frac{4}{3} \pi r^3 + 4\pi r^3 x}{r(r - 2m)}$$

$$= -\frac{4}{3} \pi r^2 (1 + x) \rho_0 \frac{1 + 3x}{r(r - 2m)}$$

$$= -\frac{4}{3} \pi r^3 (1 + x)(1 + 3x) \rho_0 \frac{1}{r} \frac{1}{r - \frac{8}{3} \pi r^3 \rho_0}$$

$$= -(1 + x)(1 + 3x) \frac{4\pi \rho_0 r}{3 - 8\pi \rho_0 r^3}$$

c)* Integrate the previous ODE while imposing $p(R) = 0$ and $p(0) = p_c$ to find

$$p_c = \rho_0 \frac{1 - \sqrt{1 - 2M/R}}{3\sqrt{1 - 2M/R} - 1}.$$

Boundary conditions mean $x(R) = 0$ and $x(0) = \frac{p_c}{\rho_0}$

$$\frac{dx}{dr} = -(1+x)(1+3x) \frac{4\pi \rho_0 r}{3 - 8\pi \rho_0 r^2}$$

$$\int_{\frac{p_c}{\rho_0}}^0 \frac{dx}{(1+x)(1+3x)} = \int_0^R \frac{4\pi \rho_0 r}{3 - 8\pi \rho_0 r^2}$$

```

In[2]: Integrate[1 / ((1 + x) (1 + 3 x)), {x, 0, 0}]
Out[2]: -1/2 (-Log[1 + 0] - Log[1 + 3 0]) If Re[0] > 0 || 1/3 < Re[0] < 0 || 0 ∈ R
In[3]: Assuming[0 > 0 && R > 0, Integrate[4 π ρ0 r / (3 - 8 π ρ0 r^2), {r, 0, R}]]
Out[3]: -1/4 Log[1 - 8/3 π R^2 ρ0] If 8 π R^2 ρ0 < 3

```

(all conditions can be easily seen to be satisfied)

$$\frac{1}{2} \left[-\ln\left(1 + \frac{p_c}{\rho_0}\right) + \ln\left(1 + \frac{3p_c}{\rho_0}\right) \right] = -\frac{1}{4} \ln\left[1 - \frac{8}{3} \pi R^2 \rho_0\right]$$

$$\frac{1}{2} \ln \frac{1 + \frac{3p_c}{\rho_0}}{1 + \frac{p_c}{\rho_0}} = -\frac{1}{4} \ln\left[1 - \frac{8}{3} \pi R^2 \rho_0\right]$$

$$\left[\frac{1 + \frac{3p_c}{\rho_0}}{1 + \frac{p_c}{\rho_0}} \right]^{1/2} = \left(1 - \frac{8}{3} \pi R^2 \rho_0\right)^{-1/4}$$

Take to the 4th power & solve

$$\text{Solve}\left[\frac{\left(1 + \frac{3p_c}{\rho_0}\right)^2}{\left(1 + \frac{p_c}{\rho_0}\right)^2} == \frac{1}{1 - \frac{8}{3} \pi R^2 \rho_0}, p_c\right]$$

$$\left\{ \left\{ p_c \rightarrow \frac{3 \rho_0 - 12 \pi R^2 \rho_0^2 - \sqrt{3} \sqrt{3 \rho_0^2 - 8 \pi R^2 \rho_0^3}}{12 (-1 + 3 \pi R^2 \rho_0)} \right\}, \left\{ p_c \rightarrow \frac{3 \rho_0 - 12 \pi R^2 \rho_0^2 + \sqrt{3} \sqrt{3 \rho_0^2 - 8 \pi R^2 \rho_0^3}}{12 (-1 + 3 \pi R^2 \rho_0)} \right\} \right\}$$

$$\rho_c = \frac{3\rho_0 - 12\pi R^2 \rho_0^2 - \sqrt{3}\sqrt{3\rho_0^2 - 8\pi R^2 \rho_0^3}}{12(-1 + 3\pi R^2 \rho_0)}$$

$$\text{Nok } \frac{4}{3}\pi R^3 \rho_0 = M$$

$$\frac{\rho_c}{\rho_0} = \frac{3 - 12\pi R^2 \rho_0 - \sqrt{3}\sqrt{3 - 8\pi R^2 \rho_0}}{12(-1 + 3\pi R^2 \rho_0)}$$

$$= \frac{3 - 9\frac{M}{R} - \sqrt{3}\sqrt{3 - 6\frac{M}{R}}}{12(-1 + \frac{9}{4}\frac{M}{R})}$$

$$= \frac{1 - \frac{3M}{R} - \sqrt{1 - 2\frac{M}{R}}}{(-4 + 9\frac{M}{R})}$$

$$\text{Define } s = \sqrt{1 - 2\frac{M}{R}}$$

$$s^2 = 1 - 2\frac{M}{R}, \frac{M}{R} = \frac{1-s^2}{2}$$

$$\frac{\rho_c}{\rho_0} = \frac{1 - \frac{3}{2}(1-s^2) - s}{-4 + \frac{9}{2}(1-s^2)}$$

$$= \frac{\frac{3}{2}s^2 - s - \frac{1}{2}}{-\frac{9}{2}s^2 + \frac{1}{2}}$$

$$= \frac{3s^2 - 2s - 1}{-9s^2 + 1}$$

$$= \frac{(1+3s)(s-1)}{(1-3s)(1+3s)}$$

$$= \frac{s-1}{1-3s} = \frac{1-s}{3s-1}$$

$$= \frac{1 - \sqrt{1 - 2\frac{M}{R}}}{3\sqrt{1 - 2\frac{M}{R}} - 1}$$

as desired

d)* Impose $\rho_c < \infty$ to find Buchdahl's inequality for constant density stars,

$$\frac{M}{R} < \frac{4}{9}$$

Show that constant density stars can get arbitrarily close to saturating this bound.

The only thing that can force ρ to blow up is the denominator.

Thus we require

$$3\sqrt{1 - 2\frac{M}{R}} - 1 > 0$$

$$3\sqrt{1 - 2\frac{M}{R}} > 1$$

$$1 - 2\frac{M}{R} > \frac{1}{9}$$

$$\frac{8}{9} > 2\frac{M}{R}$$

$\frac{M}{R} < \frac{4}{9}$, which was also clear from the expression earlier

$$\frac{\rho_c}{\rho_0} = \frac{1 - \frac{3M}{R} - \sqrt{1 - 2\frac{M}{R}}}{(-4 + 9\frac{M}{R})}$$